REFERENCE · R9

# VEHICLE RECOVERY GUIDE

Self and assisted recovery across sand, mud, rock & steep terrain

2009 LC76 SW · 1HZ 4.2L Diesel · low-boost turbo (0.6 bar) · intercooler
Last updated: 12 October 2026

| | | | |
|---|---|---|---|
| **3,000 KG** GROSS VEHICLE MASS | **9,500+ LB** MIN WINCH RATING | **4.75T** MIN SHACKLE WLL | **R14** TYRE PRESSURES |

## 1. INTRODUCTION & SAFETY PRINCIPLES

### ❌ RECOVERY SAFETY — READ FIRST

*Recovery operations are high-risk. Equipment failure can cause serious injury or death.*

❌ **NEVER** use tow balls for recovery — they become lethal projectiles
❌ **NEVER** stand in the recovery line (front or rear of vehicles)
❌ **NEVER** use damaged, frayed, or kinked equipment
❌ **NEVER** exceed equipment working load limits
❌ **NEVER** rush — assess the situation completely first

✓ **ALWAYS** use rated recovery points (ARB or equivalent)
✓ **ALWAYS** use a winch dampener on the line
✓ **ALWAYS** keep bystanders at least 1.5× rope length away
✓ **ALWAYS** wear gloves when handling ropes and cables

### 🎯 RECOVERY DECISION HIERARCHY

*Work through this list in order. Stop at the first option that works.*

1. Self-recovery without equipment — reverse out, different line
2. Reduce tyre pressure — increases footprint, improves traction
3. Use recovery boards — traction under wheels
4. Dig out / clear obstacles — shovel, pick, jack base plate
5. Vehicle-to-vehicle recovery — kinetic rope / snatch strap
6. Winch recovery — solo or assisted
7. Call for professional recovery

### 🧠 PRE-RECOVERY RISK ASSESSMENT

*Before any recovery attempt, assess these factors.*

**Terrain stability:** Is the ground getting softer? Rising water?

**Vehicle position:** Level, tilted, risk of rollover?

**Anchor points:** What's available? Trees, rocks, buried spare?

**Equipment condition:** Inspect all gear before use

**Weather:** Changing conditions? Approaching darkness?

**Personnel:** Who knows recovery procedures? Who operates controls?

*If in doubt — wait for help. A stuck vehicle is recoverable; injuries may not be.*

### 🔍 WHY YOU'RE STUCK — LIFT AND PULL

*Every bogging is a shortage of lift, of pull, or both (Sheppard 2008, §5.4-3).*

**Lift** gets the wheels back onto something firm — dig, jack, boards, lower pressure, offload. **Pull** moves the vehicle once they are — its own drive, people pushing, another vehicle, the winch. Work out which you are short of before you start.

1. Stop the wheels the moment you are stuck — every extra turn digs you deeper
2. Try reversing out first, in your own tracks — the way you came in held you (Sheppard 2008, §5.4-4)
3. Are the tyre pressures right for this ground? If not, fix them first (R14)
4. Is any weight on the chassis, diffs or sills instead of the wheels? Dig or jack that clear before anything else
5. Look at all four wheels and find the one that is holding you — work on that one first, then any other wheel without a clear path ahead
6. Finish the job before you try — a half-hearted dig that fails means doing it all again

7. Use what is around you: firm ground behind or ahead, a slight downslope, offloading heavy cargo

*Steps 1 and 3–7: St Pierre White 2014, p. 214. If the first attempts fail, stop, have a drink, and work out why before trying again.*

## 2. TERRAIN-SPECIFIC RECOVERY

### 🏜️ SAND RECOVERY

*Kalahari, Namib, coastal dunes. The 1HZ's torque works well at low RPM but lacks power for steep dunes.*

R14
SAND (SOFT, DEEP) ROW

**Recovery procedure:**

1. Stop immediately when wheels spin — don't dig deeper
2. Drop to R14's soft-sand pressures; go to its dune row only if still sinking — never below it
3. Clear sand from around tyres, underneath diffs
4. Place recovery boards in front of driven wheels
5. Select low range, diff-lock engaged
6. Drive smoothly onto boards, maintain momentum
7. Once clear, stop and retrieve boards

**1HZ Note:** Without turbo boost at low RPM, the naturally aspirated feel requires momentum planning. Don't expect to power through — use technique.

### 🌧️ MUD RECOVERY

*Black cotton soil, vleis, wet tracks. Can be deceptively deep and grab axles.*

R14
MUD ROW

**Before entering:**

• Gauge depth with a stick — look for submerged ruts, obstacles

• Check exit route — is it worse than entry?

**Recovery procedure:**

1. Engage low range, diff-lock, before getting stuck
2. If stuck, DON'T spin wheels — worsens situation
3. Try rocking: forward/reverse in quick succession
4. Clear mud from around tyres, under chassis
5. Use boards, branches, rocks under wheels for traction
6. If deep, jack the vehicle on a base plate (ARB hydraulic or bottle jack), pack material underneath

⚠️ **Mud suction can hold 3+ tonnes. Winch slowly, check for movement, don't shock-load.**

### 🪨 ROCK & TECHNICAL RECOVERY

*Hi-centering, wheel wedging, body contact. The LC76's solid axles articulate well but watch for body damage.*

R14
ROCKY TERRAIN (CRAWL) ROW

**Common situations:**

**Hi-centered:** Vehicle resting on chassis/diff, wheels off ground

• Use the ARB hydraulic or bottle jack on a base plate to raise the vehicle

• Stack rocks/boards under lifted wheel

• Lower jack, drive off obstacle

**Wedged wheel:** Tyre trapped between rocks

• DON'T force — will damage tyre sidewall

• Jack up, clear rocks, lower back

**Spotting protocol:**

• Spotter walks ahead, maintains eye contact

• Hand signals: stop, go slow, left, right, reverse

• Driver follows spotter's commands exactly

### ⛰️ STEEP HILL RECOVERY

*Mountain passes, dongas. Risk of rollover if recovery angle is wrong.*

**Stalled on ascent:**

1. Hold hard on the footbrake (no servo with the engine off). Never press the clutch to stop a stall.
2. Select low-range reverse, clutch fully out
3. Ease off the brake, then turn the key — the 76 cranks in gear and backs down under engine braking
4. Wheels straight, both hands on the wheel — turning causes rollover. Full procedure: R28 §8

**Failed descent (too steep/slippery):**

1. Stop, engage handbrake
2. Winch to anchor point UPHILL
3. Control descent with winch tension + brakes

⚠️ **NEVER turn sideways on a slope. If sliding sideways, straighten wheels and try to regain forward/backward control.**

## 3. WINCHING PROCEDURES

### 🏗️ WINCH SYSTEM OVERVIEW — LC76

*Sizing rule: at least the gross vehicle weight plus about 15% (Allen 2021, p. 203). The fitted 9,500 lb winch (~4,309 kg) covers that on its first layer, even at the measured full-trim mass. Whether a particular pull needs the snatch block is a matter of ground and slope — see the load table below.*

| COMPONENT | SPECIFICATION | NOTES |
|---|---|---|
| Winch capacity | 9,500 lb (fitted) | Rule: gross vehicle weight + about 15% (Allen 2021, p. 203). The maker's pull chart governs once in hand (OI-116) |
| Rope type | Synthetic (fitted) | Light, little stored energy if it breaks. Keep at least 8 wraps on the drum (wire needs 5); keep it off sharp edges; avoid powering out or lowering a vehicle under load — the drum brake heats the rope (Allen 2021, ch. 8) |
| Rope length | 30m minimum | More line = weaker pull; use snatch block for distance |
| Fairlead type | Hawse (synthetic) / Roller (steel) | Match fairlead to rope type |
| Line speed | Layer 1 = slowest/strongest | Always use lowest layer when max pull needed |

**Electrical doctrine while winching:** the winch is wired **crank-direct** — it draws off the Enertec 650B-BLS crank battery (several hundred amps under load), not the LiFePO4 house bank (R26 §1 wiring note). ALWAYS winch with the **engine running above idle** so the alternator supports the crank battery (the T-Max figure for rpm and current is being confirmed — OI-116); after a long pull, leave the engine running so the battery recovers before you switch off (Allen 2021, ch. 8); a stalled pull on a dead engine can flatten the crank battery in minutes and leave you stuck AND unable to start. Winch in **bursts with cooling pauses** — winch motors are short-duty-cycle devices, and the motor runs hotter as the battery voltage falls. Stop periodically to let it cool and the battery recover; keep the voltage above about 10 V (Allen 2021, p. 203). The maker's duty cycle governs once in hand (OI-116). The M-series switch and house electronics play no part in winching.

**Pull rating drops with every layer of rope on the drum:** the rated pull is on the first layer; each layer after that gives roughly 10% less (Allen 2021, p. 211). For a hard pull, spool out until you are on the bottom layer, or double-line. The T-Max chart governs once in hand (OI-116).

### 🌳 ANCHOR SELECTION

*A failed anchor can become a projectile. Always verify before loading.*

**Trees:**

• A live, sound tree — the thicker the better; if in doubt, use two trees or another anchor

• Use tree protector strap — wide, no metal

• Wrap LOW on trunk (strongest point)

• Avoid dead trees, leaning trees, shallow-rooted species

**Rocks/Boulders:**

• Must be solidly embedded, not sitting on surface

• Wrap strap around, not hooked over edge

**Buried spare wheel (land anchor):**

• Dig a hole at least as deep as the tyre is tall, with a sloping slit trench toward the vehicle for the strap; keep the side facing the pull vertical and solid (Allen 2021, p. 217)

• Bury wheel flat, attach strap through centre

• Pack firmly, wet if possible

**Another vehicle:**

• Must be secure — brakes on, wheels chocked

• Connect to rated recovery points ONLY

### ⚙️ WINCHING SETUP PROCEDURE

*Proper setup prevents equipment damage and injury.*

1. Stuck vehicle: engine running, low range, driver at the wheel. Bonnet up to shield the windscreen. One person only works the winch control (St Pierre White 2014, p. 216, 220)
2. Disengage winch clutch (free-spool mode)
3. Walk rope to anchor — don't drive over it
4. Wrap tree protector low on tree trunk
5. Connect shackle through strap eyes (pin finger-tight + ¼ turn back)
6. Connect winch hook or soft shackle to strap shackle
7. Engage winch clutch, take up slack SLOWLY
8. Place winch dampener on rope (heavy blanket/bag works)
9. Clear ALL personnel to 1.5× rope length
10. Operator: signal ready, begin winching in short pulls. Stuck vehicle drives gently in low-range second, matching the winch speed — no wheelspin; if it lurches and jerks the line, go to neutral and let the winch work (Allen 2021, ch. 8)

### 🔄 SINGLE VS DOUBLE LINE PULL

*Double-line using snatch block doubles pulling power but halves speed.*

**Single line:**

• Direct connection, full speed

• Use when pull required ≤ winch capacity

**Double line (snatch block):**

• Attach snatch block to anchor

• Run winch rope through block, back to vehicle recovery point

• Effective pull = 2× winch rating (minus friction)

• Speed = 50% of single line

• Required for heavy mud suction, steep inclines

**When to double-line:** use the load table below.

### 📐 HOW HARD IS THE PULL? — LOAD TABLE

*Rough pull needed, so you know before you start whether to double-line (Allen 2021, p. 216).*

**Method:** pull ≈ (slope in degrees ÷ 60 × loaded mass) + (ground share × loaded mass). Ground shares are averages from winch makers: pavement 2–4%, grass 8–14%, hard-packed sand 10–17%, wet sand 15–20%, gravel 10–20%, soft dry sand 25–30%, shallow mud 30–35%, deep mud 40–60%, deep clay mud 50–70%. The table uses the top of each range.

| GROUND | LEVEL | 10° | 20° | 30° |
|---|---|---|---|---|
| Hard sand | 560 | 1,100 | 1,650 | 2,200 |
| Gravel / wet sand | 660 | 1,200 | 1,750 | 2,300 |
| Soft dry sand | 980 | 1,530 | 2,080 | 2,620 |
| Shallow mud | 1,150 | 1,690 | 2,240 | 2,790 |
| Deep mud | 1,970 | 2,510 | 3,060 | 3,610 |
| Deep clay mud | 2,300 | 2,840 | 3,390 | 3,940 |

Kilograms of pull, worked at the measured full-trim mass of 3,280 kg (Facts §1) — rework them if the loaded mass changes. Compare with the winch: 9,500 lb (~4,309 kg) on the first layer, about 10% less per layer after that.

**When to double-line:** when the figure for your ground and slope comes near the winch's rating for the layer you are on; when the vehicle is bellied (the table assumes it is on its wheels); in suction mud, which can exceed these averages; or when a single-line pull has not moved it. Allen's worked example applies the ground share to the slope figure; the library applies it to the vehicle's mass, as his own table is headed — otherwise level ground would need no pull at all.

## 4. RECOVERY EQUIPMENT CHECKLIST

### ✅ LC76 RECOVERY KIT — RATED FOR 3,000KG GVM

*All equipment must be rated. Never substitute hardware store items for recovery gear.*

| ITEM | SPECIFICATION | QTY | CHECK |
|---|---|---|---|
| Winch | 9,500–12,000 lb, synthetic rope | 1 | ☐ |
| Bow shackles | 4.75-ton WLL minimum, rated | 4 | ☐ |

| ITEM | SPECIFICATION | QTY | CHECK |
|---|---|---|---|
| Soft shackles | 14,000 kg MBS | 2 | ☐ |
| Tree protector strap | 8-ton, 3m length | 2 | ☐ |
| Kinetic recovery rope | 8,000 kg MBS, 9m length | 1 | ☐ |
| Snatch strap | 8,000 kg, 9m (backup) | 1 | ☐ |
| Static tow rope | Ordinary towing only — never for snatching | 1 | ☐ |
| Snatch block | 10-ton capacity | 1 | ☐ |
| Winch extension strap | 10-ton, 20m | 1 | ☐ |
| Winch dampener | Purpose-built or heavy blanket | 1 | ☐ |
| Recovery boards | MAXTRAX or equivalent | 2–4 | ☐ |
| ARB hydraulic jack | Primary lift — rated for GVM | 1 | ☐ |
| Bottle jack | Backup lift / backup stand / bead breaking (R40 §3) | 1 | ☐ |
| Jack base plate / board | Prevents sinking in soft ground | 1 | ☐ |
| Shovel | Long-handle, D-grip | 1 | ☐ |
| Air compressor | Portable, 12V, 10+ bar | 1 | ☐ |
| Tyre deflators | Rapid deflation type | Set | ☐ |
| Tyre pressure gauge | Accurate to low pressure | 1 | ☐ |
| Work gloves | Leather, heavy-duty | 2 pairs | ☐ |
| Ground sheet | For lying on wet/dirty ground | 1 | ☐ |

⚠️ **Inspect all equipment before each trip. Replace any damaged, frayed, or corroded items.**

## 5. VEHICLE-TO-VEHICLE RECOVERY

❌ **CRITICAL: NEVER USE TOW BALLS**

*Tow balls are NOT rated recovery points. They can shear and become 3kg projectiles at 100+ km/h.*

❌ Tow balls are designed for STATIC towing loads

❌ Kinetic recovery loads exceed tow ball ratings by 500%+

❌ Ball shearing has caused fatalities

✓ **USE ONLY:** Factory recovery points or aftermarket rated recovery points (ARB, KAON, etc.)

🔗 **KINETIC ROPE RECOVERY**

*Kinetic (snatch) ropes and straps stretch roughly 15–30% depending on the make, storing energy that pulls the stuck vehicle out (Allen 2021, pp. 186, 188; Cobley 2013, ch. 15).*

1. Position recovery vehicle on firm ground, in line with stuck vehicle
2. Connect kinetic rope to RATED recovery points on both vehicles
3. Lay out the rope with about 1–2 m of slack, no knots or twists (Cobley 2013, ch. 15; owner ruling, 12 Oct 2026). St Pierre White has the tug stop no closer than half the rope's length (2014, p. 221). Damper or blanket on the rope
4. Clear all personnel to side, 1.5× rope length away
5. Stuck vehicle: low gear, handbrake OFF, driver applies gentle power on the signal — just short of wheelspin (Allen 2021, ch. 8; Cobley 2013, ch. 15)
6. Recovery vehicle: pull away in first at normal take-off speed, accelerate very gently and hold the speed constant — the rope's stretch does the work, not engine power. More speed does not mean more pull (St Pierre White 2014, pp. 221–222)
7. Continue smoothly — rope stretches, stores energy, pulls
8. Once stuck vehicle moves, both vehicles proceed together
9. Stop on firm ground, disconnect, inspect rope

⚠️ **Never shock-load a static rope (tow strap). Only kinetic ropes are designed for snatch recovery.**

⚠️ **Don't snatch a vehicle that is bellied — dig or jack the weight back onto the wheels first. The tug should be similar in size and weight to the stuck vehicle: a light tug recoils into it, a heavy one can tear it apart (St Pierre White 2014, pp. 221–222). Use both recovery points with a bridle where fitted (Cobley 2013, ch. 15).**

⚠️ **Joining two straps or ropes: never with a shackle — if one breaks, the shackle becomes a missile. Pass the eye of one through the eye of the other, then the far end of the second back through the first eye, and pull up with a stick or rolled cloth in the join so it can be undone. Keep the pull within about 10° of straight; recovery speed no more than a fast walk (St Pierre White 2014, p. 221–222; Allen 2021, ch. 8).**

**Rope care:** measure the rope's length when new and retire it from snatching once it has stretched 10% beyond that — keep it for towing. A wet rope is 10–15% weaker; sunlight ages it; wash out grit after use (St Pierre White 2014, p. 222; Allen 2021, ch. 8).

🪢 **LONG-ROPE AND TANDEM TOW**

*Two or more vehicles pulling together, under one marshaller (Sheppard 2008, §5.4-10).*

A long static tow rope — 30–50 m — keeps the towing vehicle off the soft patch that stopped the first one. No rope that long is in the kit — the static tow rope carried is for an ordinary tow — so this is for travelling with others or using theirs.

1. One marshaller, 15–30 m ahead of the lead vehicle, where every driver can see them. Brief strangers on the signals first
2. Marshaller waves each vehicle forward in turn until the ropes are tight
3. Hand circling: everyone in gear, revs up
4. Hand drops: all clutches in together — the stuck vehicle drives too
5. Stop signal once the vehicle is clear; tug backs a metre to take the tension off before unhooking

Towing out backwards, a second marshaller relays signals to the driver facing away. Keep everyone clear of a tensioned rope (Sheppard 2008, §5.4-10).

### ✋ HAND SIGNALS

*Clear communication prevents accidents. Agree signals before starting.*

**STOP:** Closed fist raised high

**GO:** Open palm, waving forward

**SLOW DOWN:** Palm down, pressing down motion

**REVERSE:** Beckoning toward yourself

**LEFT:** Point left with extended arm

**RIGHT:** Point right with extended arm

*If radios available, use channel agreed beforehand. Keep transmissions brief and clear.*

## 6. SELF-RECOVERY WITHOUT WINCH

### 📋 RECOVERY BOARDS

*First line of self-recovery. Work in sand, mud, and snow.*

1. Clear material from in front of tyres
2. Place boards in front of driven wheels (usually rear)
3. Angle boards down slightly into surface
4. Select low range, diff-lock if available
5. Drive SLOWLY onto boards — no wheel spin
6. Once grip established, continue smoothly off boards
7. Stop on firm ground, retrieve boards

**Boards ejecting:** If wheels spin on boards, they can shoot out dangerously. Never stand behind wheels during recovery.

### ⛏️ DIG & FILL METHOD

*When stuck to the axles, digging is mandatory before any recovery attempt.*

1. Dig channels in front of all four wheels
2. Clear material from under differentials and chassis rails
3. Fill channels with rocks, branches, boards — anything solid
4. Build a ramp up to firmer ground level
5. Reduce tyre pressure if not already done
6. Drive out slowly, using low range

*This is slow, physical work but often the only option when alone without a winch.*

### 🔧 JACK TECHNIQUES — ARB HYDRAULIC & BOTTLE

*No hi-lift is carried — deliberate doctrine. Recovery pulls belong to the winch and MaxTrax; lifting belongs to the hydraulic and bottle jacks.*

**Lifting to insert traction:**

1. Chock the opposite wheels; place the jack base plate/board — always on soft ground
2. Position the ARB hydraulic or bottle jack under the axle at a rated solid-axle jacking point (R40 §2)
3. Lift the wheel clear of the ground

4. Pack material / recovery boards under the wheel

5. Lower slowly, remove the jack, drive off

**Doctrine & deferral:** the pulling role a hi-lift would serve is owned by the winch and MaxTrax (this guide); the full jacking standard, bead breaking and all wheel/tyre spanner work are owned by R40 · Wheel & Tyre Field Procedures.

⚠️ **A jack is a lifting tool, not a stand — never work under a vehicle held on a jack alone. When lifting with the hydraulic jack, set the bottle jack under the axle as a backup stand (R40 §2).**

### 🔄 ROCKING TECHNIQUE

*Uses vehicle momentum to break free from shallow bogs.*

1. Select low range
2. Drive forward until wheels spin or vehicle stops
3. Immediately shift to reverse, back up until stopped
4. Shift forward again, using momentum from reverse
5. Repeat, building rhythm and rocking amplitude
6. On forward swing, continue out if traction gained

**Clutch warning:** Rapid forward/reverse is hard on the R151F gearbox. Don't over-rev, shift smoothly, and stop after five cycles if it is not gaining (decision tree, §8). Not in sand — rocking usually digs the vehicle in deeper there; reverse out instead (R28 §1; St Pierre White 2014, p. 183).

## 7. COMMON MISTAKES & LESSONS LEARNED

### ❌ MISTAKES THAT CAUSE DAMAGE

*Lessons from recovery failures — avoid these.*

❌ **Spinning wheels when stuck:** Digs deeper, heats tyres, wastes fuel
❌ **Recovering at wrong angle:** Side-loads damage equipment, causes rollovers
❌ **Using worn/damaged equipment:** Shackle pins can fail, straps can snap
❌ **Attaching to steering components:** Never use track rod, tie rod, steering arm
❌ **Not using winch dampener:** If cable/rope breaks, recoil can kill
❌ **Jerking with static strap:** Shock loads destroy straps and connection points
❌ **Standing in the line:** Never between vehicles or in line with rope

### ✓ BEST PRACTICES

*Habits that prevent problems and improve outcomes.*

✓ **Stop early:** Don't "try a bit more" — assess before deeper
✓ **Deflate first:** Often solves the problem without recovery gear
✓ **Dig before pulling:** Reduce resistance before applying force
✓ **Use a bridle:** Connect to two points for straight-line pull
✓ **Check equipment after each use:** Damage may not be obvious
✓ **Practice with your gear:** First use shouldn't be an emergency
✓ **Travel in convoy:** Vehicle-to-vehicle recovery is easier than solo

### 🔋 1HZ-SPECIFIC CONSIDERATIONS

*The turbocharged 1HZ has specific characteristics affecting recovery.*

**Power delivery:**

• Peak torque (373Nm after tune) available at 2,200 RPM
• SAC low-boost turbo adds ~30% power over NA spec
• Intercooler reduces intake temps — important when winching

**During winching:**

• Winch draws several hundred amps — watch the voltmeter; pause and let the battery recover if it stays low (Allen 2021, ch. 8)
• Engine running above idle while winching — T-Max figure to be confirmed (OI-116)
• Watch EGT if engine loaded during recovery (climbing)

**In sand:**

• 1HZ lacks instant throttle response — maintain momentum
• Don't expect to power out of deep sand like V8 diesels

• Technique beats power: tyre pressure and line selection are critical

## 8. QUICK REFERENCE PROCEDURES

### 📋 DECISION TREE: STUCK IN SAND

**STOP** immediately — don't dig deeper

↓ Deflate tyres to R14's soft-sand pressures

↓ Try reverse — is there grip?

→ YES: Reverse out, find new line

→ NO: Place recovery boards

↓ Drive onto boards — vehicle moving?

→ YES: Continue to firm ground

→ NO: Dig channels, repeat boards

↓ Still stuck? Winch or vehicle recovery

### 📋 DECISION TREE: STUCK IN MUD

**STOP** immediately — assess depth

↓ Engage low range, diff-lock (if not already)

↓ Try rocking forward/reverse (5 cycles max)

→ Vehicle moving? Continue out

→ No movement? Exit vehicle, assess

↓ Dig around tyres and under chassis

↓ Pack branches/rocks under wheels

↓ Place recovery boards if available

↓ Try again — moving?

→ YES: Continue to firm ground

→ NO: jack + fill, or winch/tow

### 📋 WINCH RECOVERY CHECKLIST

- [ ] Stuck vehicle: engine running, low range, driver ready to assist the pull; bonnet up
- [ ] Anchor point identified and checked (live, sound tree or solid rock)
- [ ] Tree protector wrapped LOW on trunk
- [ ] Shackles connected correctly (pin finger-tight + ¼ turn back)
- [ ] Winch clutch engaged (not free-spool)
- [ ] Winch dampener placed on rope
- [ ] All personnel cleared to 1.5× rope length
- [ ] Winching vehicle: engine running above idle (T-Max figure: OI-116); at least 8 wraps left on the drum
- [ ] Operator signaled ready
- [ ] Winch in short pulls, checking movement
- [ ] Once clear, drive to firm ground
- [ ] Inspect all equipment before storing

### 📋 KINETIC RECOVERY CHECKLIST

- [ ] Vehicle on its wheels, not bellied — dig or jack first
- [ ] Tug similar in size and weight; in line, within about 10° of straight
- [ ] Rated recovery points both ends (bridle where fitted) — never the tow ball
- [ ] Joining two ropes: eye through eye, never a shackle; stick in the join
- [ ] About 1–2 m slack, no knots; damper on the rope

- [ ] Everyone clear to the side
- [ ] Stuck vehicle in low gear, gentle power on the signal
- [ ] Tug: first gear, gentle, constant speed — no run-up
- [ ] Rope checked and length noted after use (retire from snatching at +10%)

## 📊 TYRE PRESSURE QUICK REFERENCE — LC76

| WHERE YOU ARE | R14 ROW TO USE | NOTES |
|---|---|---|
| Back on tar | Tar highway (laden or unladen) | Re-inflate before the tar — no exceptions |
| Corrugated gravel | Corrugated dirt | Keep to that row's speed ceiling |
| Beach and sand tracks | Sand (soft, deep) | Start here |
| Soft sand or dunes, still sinking | Sand (extreme — dunes) | Last resort and the floor — never lower; watch for rim damage |
| Mud | Mud | Balance between flotation and self-cleaning |
| Rock | Rocky terrain (crawl) | Protect sidewalls; avoid too soft |

R14 gives the front and rear pressure and the speed ceiling for each row; the Tyre Pressure Calculator works them out for your load.

**Always reinflate** before returning to sealed roads. Driving deflated on tar destroys tyres and causes blowouts.